

Bioinformatics and Public Health Informatics

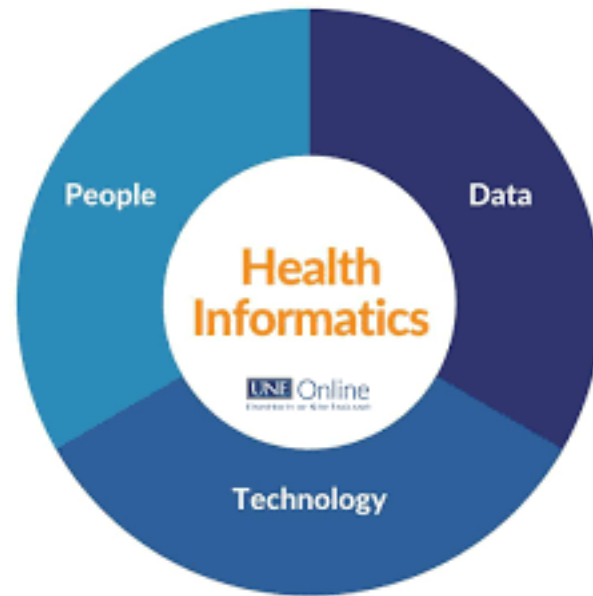
Reference: Chapter 18 & 19 Robert E. Hoyt. Medical informatics-Practical guide for health care professionals,
3rd Edition, 2009

Learning outcomes

- ▶ Describe Bioinformatics and its interface with medical informatics
- ▶ List various databases for Bioinformatics

What is Health informatics

- ▶ Health informatics saves lives by combining information technology and computer science with clinical expertise and management for better patient care.

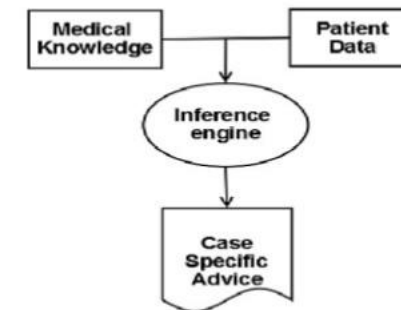


Fundamentals of Health Informatics

- ▶ Electronic prescribing.
- ▶ Information Retrieval.
- ▶ Clinical Decision Support System.

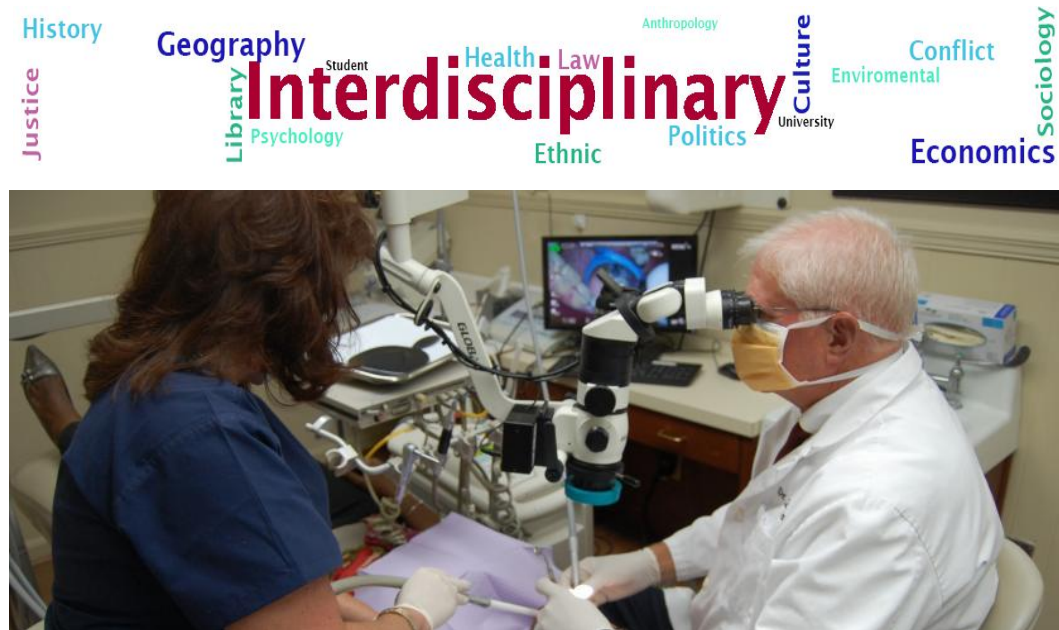


Clinical Decision Support



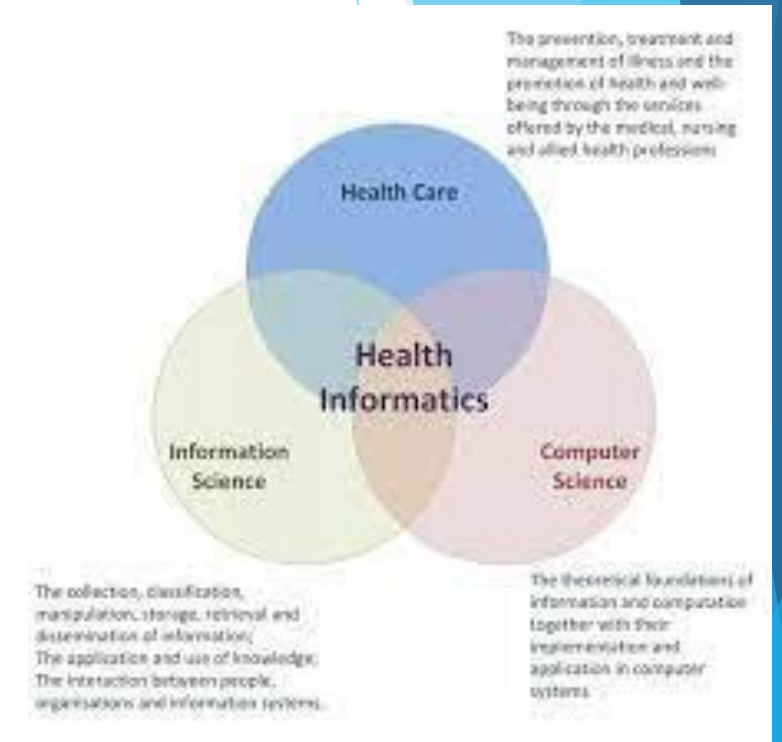
Implementation Strategy

- ▶ Interdisciplinary Course / Elective Course.
- ▶ Curriculum for Dental Researchers.



Application of Health Informatics

- ▶ Storing, grouping, and analysing large amount of health related data.
- ▶ Statistical analysis of health related data.
- ▶ Evaluation of data obtained by health surveys.
- ▶ Graphical presentation of health related data.
- ▶ Patient information, education and motivation from health related data.



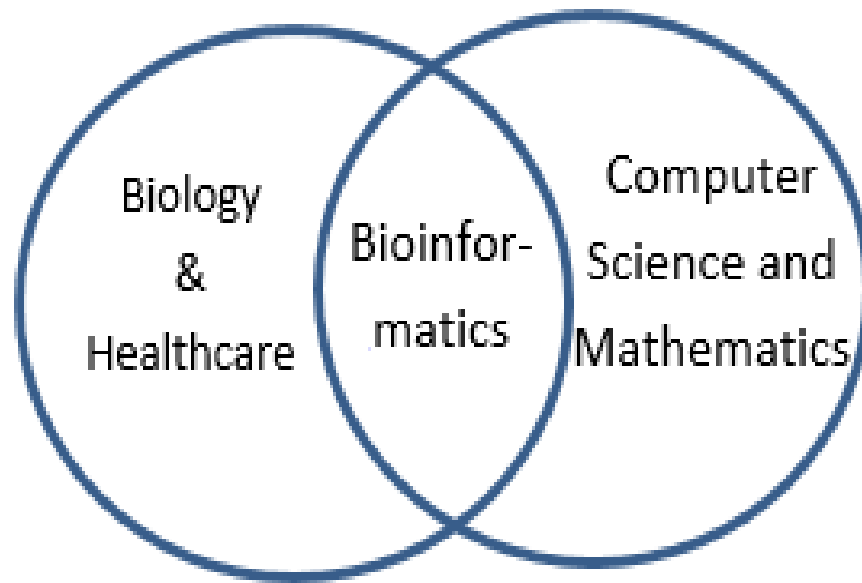
Challenges

- ▶ Secure user authentication.
- ▶ Universal access to computers and high-speed Internet connections.



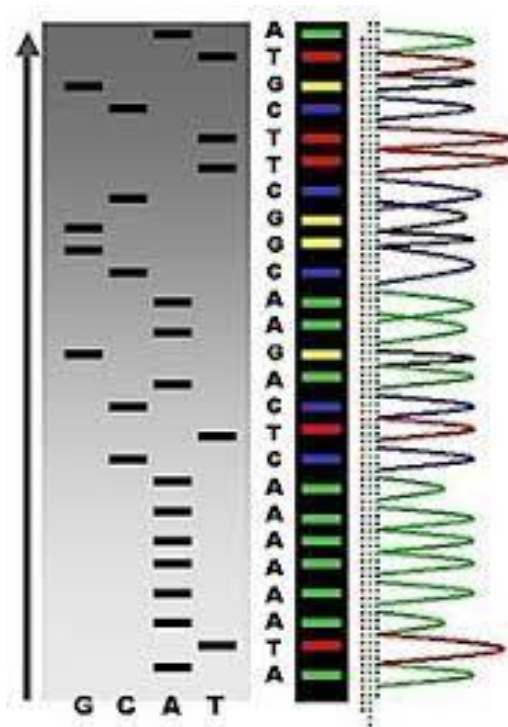
Bioinformatics

- ▶ “The field of science in which biology, computer science and information technology merge to form a single discipline”



The field involves

- ▶ Biological databases that helped evaluate DNA sequences and other genetic proteins.



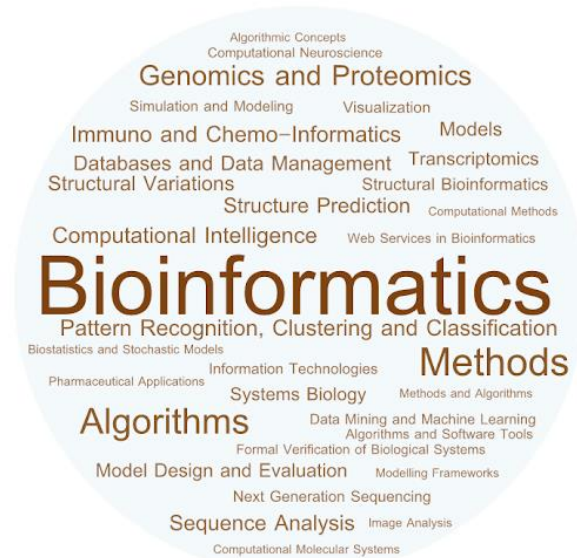
Computational biology

- ▶ The process of interpreting genetic data is referred to as computational biology.



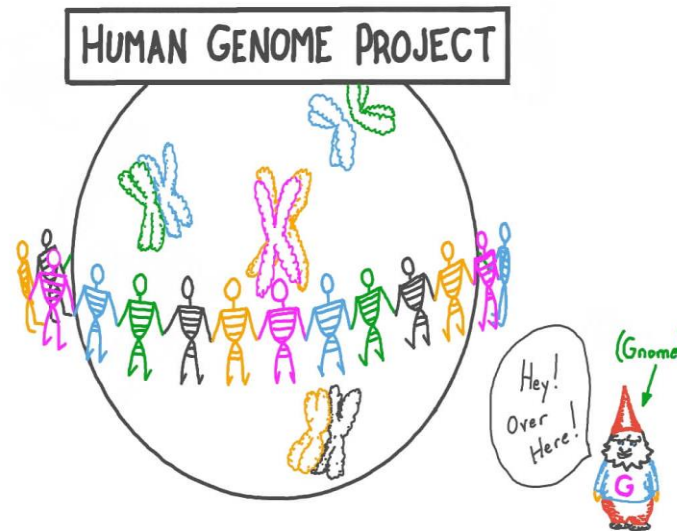
Bioinformatics terms

- ▶ Genomics: The field that analyzes genetic material from a species.
- ▶ Proteomics: The study of gene expression at the level of proteins.
- ▶ Pharmacogenomics: The study of genetic material to look for drug targets.



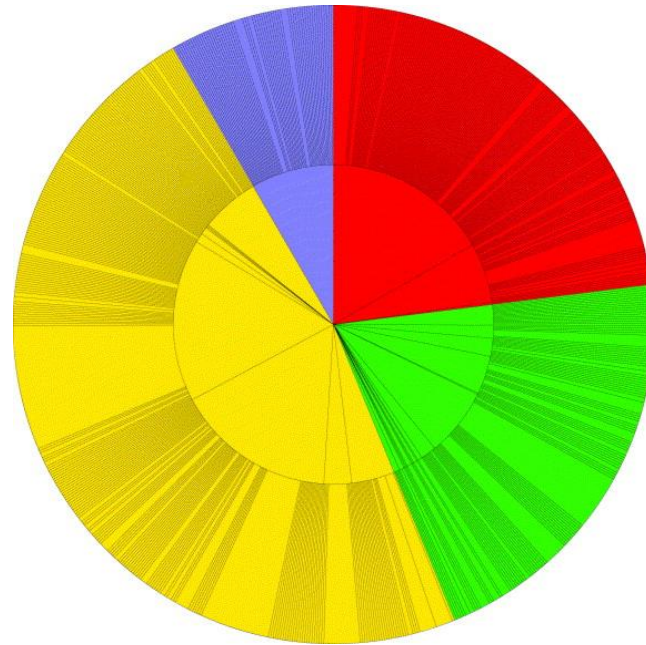
Bioinformatics Programs

- ▶ The Human Genome Project (HGP)
- ▶ Human Variome Project
- ▶ National Center for Biotechnology Information



What is the future direction of Bioinformatics?

- ▶ Integration with electronic health records
- ▶ Personal Genetics
- ▶ Population Studies



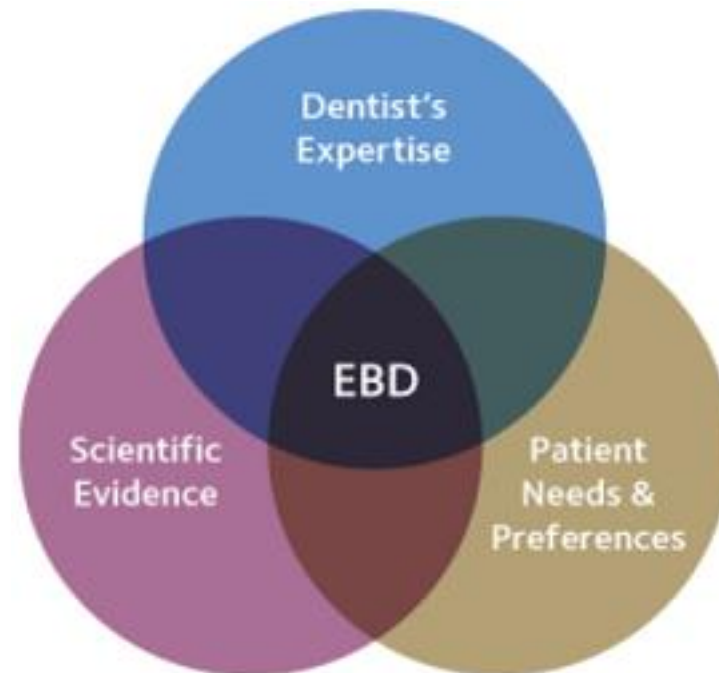
What is Dental informatics?

- ▶ Dental informatics is the application of computers and information science to improve dental research, education and management.



The reason by which oral health care be improved by Dental informatics

- ▶ Dental informatics paves the way for building up database for EBD (Evidence Based Dentistry), by which approach to oral health care can be improved.



Goals of Dental Informatics

- ▶ Improve patient outcomes.
- ▶ Delivery of dental care more competent.
- ▶ Solve clinical problems more with sound evidence based.





Introduction to Public Health Informatics

Learning Outcomes

- Define the scope and goals of public health informatics
- Describe the various disparate public health informatics programs

Topics

Introduction to Public Health Informatics

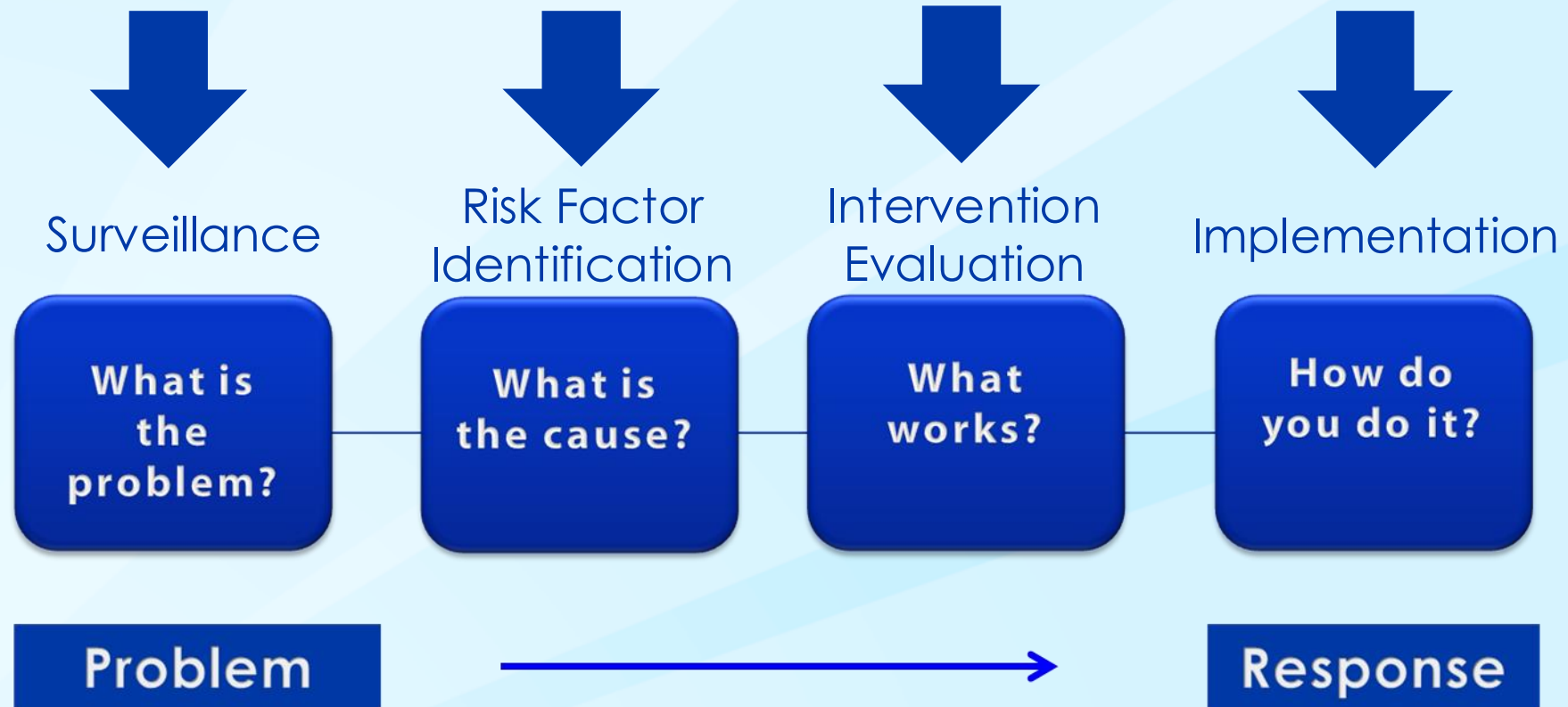
1. A Public Health Approach
2. Public Health Informatics Definition and Components
3. Creating a Public Health Information System
4. At the Intersection of the Informatician and the Information Technologist

Topic 1

A Public Health Approach



A Public Health Approach (Steps)



Topic 2

Public Health Informatics Definition and Components



Public Health Informatics — Defined



Public health informatics is the systematic application of information, computer science, and technology to public health practice, research, and learning

Yasnoff WA, O'Carroll PW, Koo D, Linkins RW, Kilbourne EM. Public health informatics: improving and transforming public health in the information age. *J Public Health Manag Pract* 2000;6:67–75.

Riegelman R, ed. *Public health 101: healthy people—healthy populations*. Sudbury, MA: Jones & Bartlett Learning; 2010: 40.

The Public Health Mission



Provide crucial scientific
information that protects our
nation against dangerous and
costly health threats

Building Your Public Health Information System (Components)



Programmer



Database Administrator



Informatician



Network Administrator



Web Designer



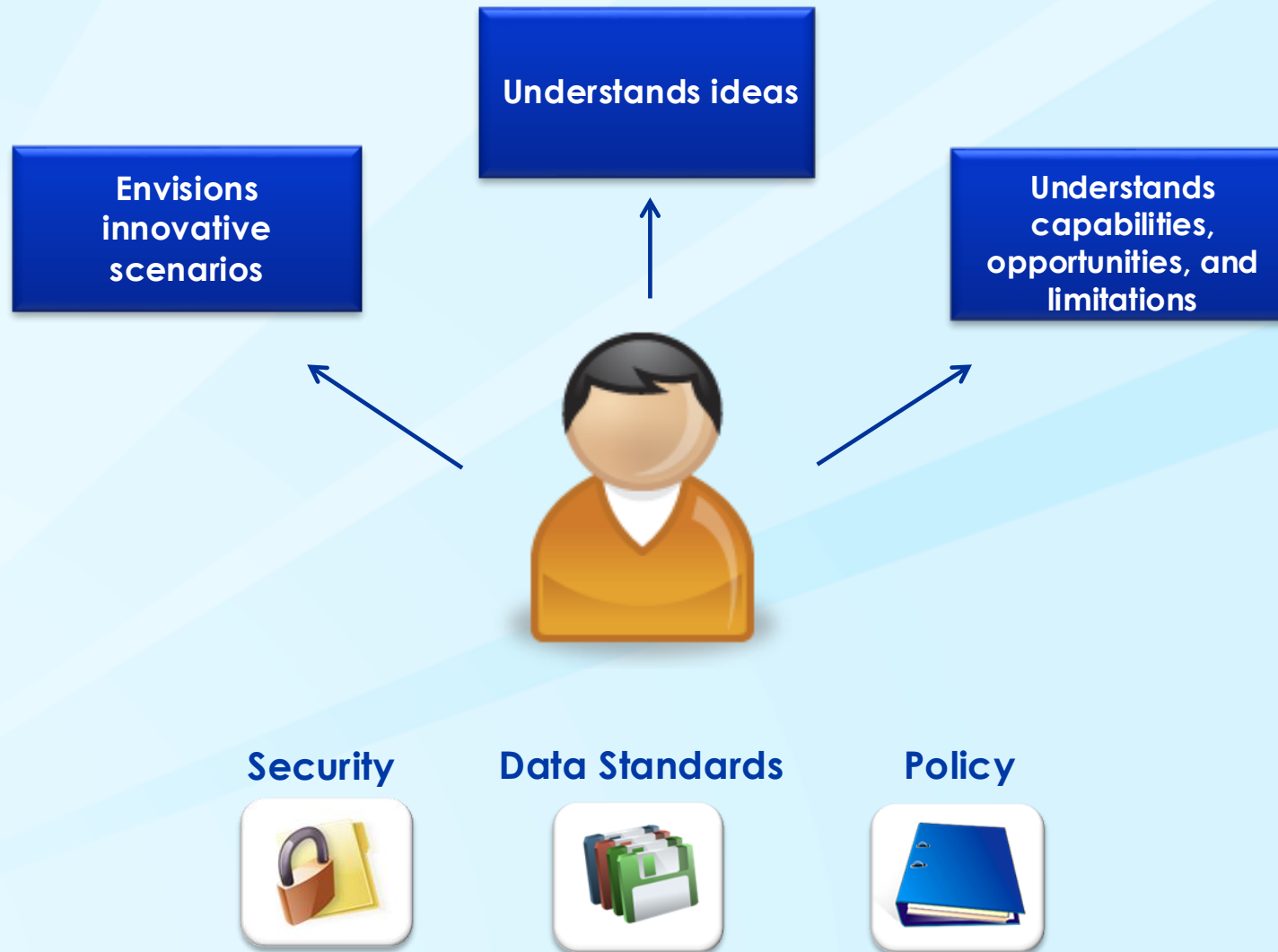
Security Specialist

Topic 3

Creating a Public Health Information System



The Informatician



Creating a Public Health Information System

Creating a Public Health Information System
Step 1 — Vision and System Planning
Step 2 — Health Data Standards and Integration
Step 3 — Data Privacy and Security
Step 4 — Systems Design and Implementation
Step 5 — Visualization, Analysis, and Reporting of Health Data

CMS = Centers for Medicare and Medicaid Services; EHR = electronic health record; ELR = electronic laboratory record; HIE = health information exchange; HIPAA = Health Insurance Portability and Accountability Act; HL7 = Health Level 7; ICD = *International Classification of Diseases*; SNOMED = Systematized Nomenclature of Human Medicine; SQL = structured query language.

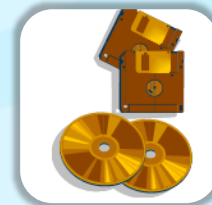
Step 1 — Vision and System Planning



Hardware



Software



Communication
Technology

Creating a public health information system

Step 2 — Health Data Standards and Integration



Is required when defining the data.

Step 3 — Data Privacy and Security



Data privacy and security must be identified, prescribed, and implemented throughout the data lifecycle.

Step 4 — Systems Design and Implementation



- Define or design methods for public health functions
- Implement/ carry out information technology for defined public health functions

Step 5 — Visualization, Analysis, and Reporting of Health Data



It is meaningful use of the data collected and managed by the system.

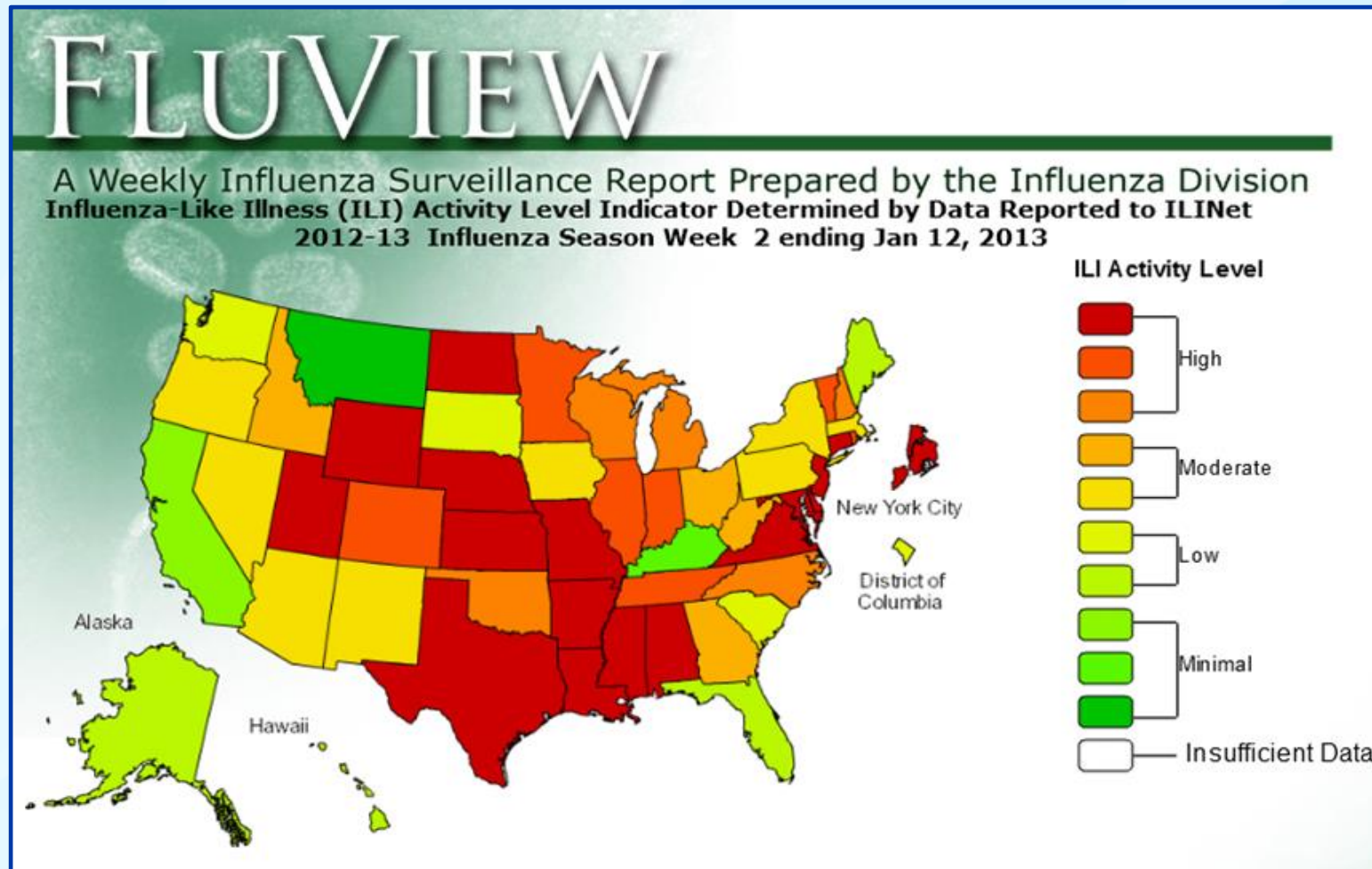
Informatics in Action



A clear-cut way to share national data is needed by whom?

1. Public health community
2. Clinicians
3. Scientists
4. General public
5. All of the above

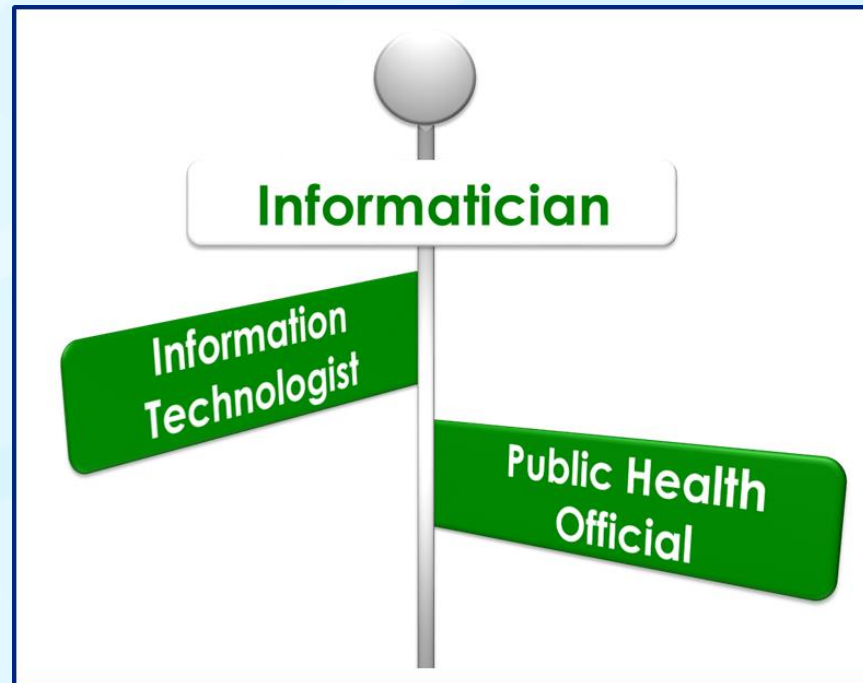
Informatics in Action — FluView



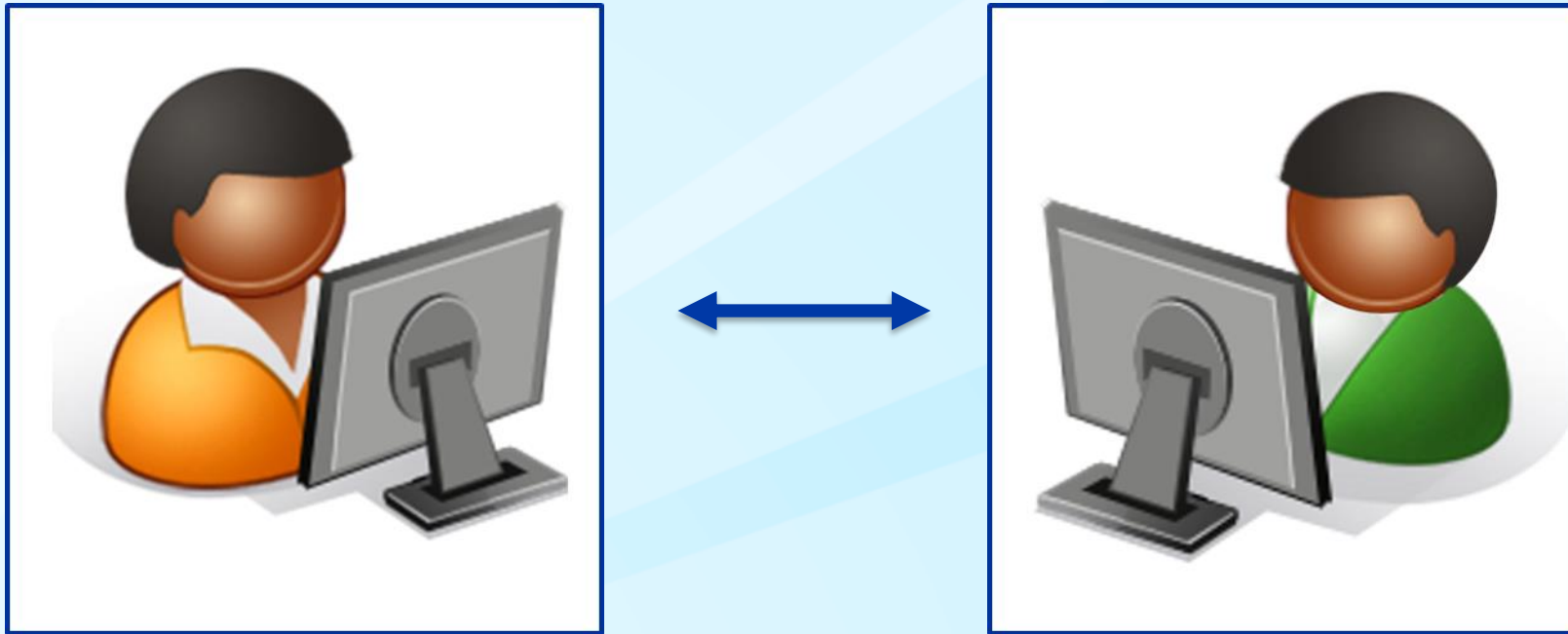
Centers for Disease Control and Prevention (CDC). FluView. Atlanta, GA: US Department of Health and Human Services, CDC; 2013. <http://gis.cdc.gov/grasp/fluview/main.html>.

Topic 4

At the Intersection of the Informatician and the Information Technologist



Combined Disciplines — The Informatician and the Information Technologist



The Role of the Informatician in Public Health



- Plans, designs, and defines functional requirements for public health information systems
- Display the data and track the progress of attaining this goal
- Evaluates the application and impact of information systems in support of health goals

The Role of the Information Technologist in Public Health



- Plans technology projects and milestones, develops software, and maintains and operates systems
- Designs, implements, and administers database architecture, privacy, security, and backup procedures



Knowledge Check

One of the United Nations' Millennium Development Goals is to substantially reduce infant mortality worldwide. A system has been developed that will display the data and track the progress of attaining this goal.

Which of the following professionals works with health data standards and sources and ensures the integrity and security of the standards?

A. The information technologist



B. The informatician

Thank you